Dashboard / Meine Kurse / 2023S365249 / Final Exam 29th June -- 17:15 -- 21:00 / [Exam] Imitation Learning

Begonnen am	Thursday, 23. March 2023, 17:19
Status	Beendet
Beendet am	Thursday, 23. March 2023, 17:29
Verbrauchte Zeit	9 Minuten 53 Sekunden
Bewertung	**4,83** von 10,00 (**48,33**%)

Frage **1**

Vollständig

Erreichte Punkte 1,00 von 1,00

What is the goal of Imitation Learning?

Wählen Sie eine oder mehrere Antworten:

- [x] a. Given a collection of episodes generated by an expert, or the expert itself, train a policy to produce the expert demonstrations
- [ ] b. Given a collection of episodes generated by an expert, or the expert itself, train a policy to maximize the future expected reward
- [ ] c. Given a collection of demonstrations or a demonstrator, train a policy to be equal or better than the demonstrator in terms of average score.
- [ ] d. Given an environment, a policy class and a loss function, train a policy to mimic the optimal policy

Frage **2**

Vollständig

Erreichte Punkte 0,50 von 1,00

What is/are the limitations of Behavioral Cloning?

Wählen Sie eine oder mehrere Antworten:

- [ ] a. It is limited only to function approximation (i.e. Neural Networks), where generalization over states is possible.
- [x] b. You need the states and the expert's actions along the trajectory.
- [ ] c. You will have to deal with distribution shift produced by accumulation of compounding errors
- [ ] d. It is limited to the tabular case, where every state has its own predictor.
- [ ] e. It is limited to Markov Decision Processes (MDPs), where all the information you need to predict the action is encoded in the current state.

Frage **3**

Vollständig

Erreichte Punkte 0,50 von 1,00

Why does causal confusion happen in Imitation Learning?

Wählen Sie eine oder mehrere Antworten:

- [x] a. Because Supervised Learning learns correlations which does not necessarily imply causation.
- [ ] b. Causal confusion only happens when using Behavioral Cloning in Partial Observable Markov Decision Processes (POMDPs)
- [ ] c. Because the training procedure (Supervised Learning) is unaware of the causal structure on the interaction between the policy and the environment.
- [ ] d. Causal confusion only happens in Behavioral Cloning
- [ ] e. Because the expert is not always selecting the same action in a given state (i.e. irrelevant actions, different strategies, demonstrations from different experts)

Frage **4**

Vollständig

Erreichte Punkte 0,67 von 1,00

Under what circumstances, Interactive Expert methods (as studied in class) are possible?

Wählen Sie eine oder mehrere Antworten:

- [x] a. When we have enough data to use Neural Networks
- [ ] b. When we have the model of the environment $p(s' \mid s, a)$
- [x] c. When we have an expert that we can ask
- [ ] d. When the state and action space is small enough to use a table (tabular case)

Frage **5**

Vollständig

Erreichte Punkte 1,00 von 1,00

What are the benefits of Imitation Learning in Reinforcement Learning (in general terms)?

Wählen Sie eine oder mehrere Antworten:

- [x] a. It gives you a good initialization of your agent
- [ ] b. It gives you the optimal policy almost for free
- [x] c. It helps with exploration in high dimensional state and action spaces
- [ ] d. It is an valid alternative to get the optimal policy
- [x] e. It is very easy to implement
- [x] f. It is very useful when rewards are sparse
- [ ] g. It is a powerful way to overcome catastrophic forgetting

Frage **6**

Vollständig

Erreichte Punkte 0,17 von 1,00

What were the problems that ALVINN faced back in the late 80's?

Wählen Sie eine oder mehrere Antworten:

- [x] a. The network was so small that it was forgetting past non i.i.d data (catastrophic forgetting)
- [x] b. Back in this time there was not computational power enough to process data "on-the-fly"
- [ ] c. Too much variance in the training samples
- [x] d. When training "on-the-fly" the data was not i.i.d and led to catastrophic forgetting
- [ ] e. Not enought variance in the training samples
- [ ] f. Same distributions in training and test data
- [x] g. Different distributions in training and test data

Frage **7**

Vollständig

Erreichte Punkte 1,00 von 1,00

What of the following statements are TRUE regarding the conection between Imitation Learning (IL) and Inverse Reinforcement Learning (IRL)

Wählen Sie eine oder mehrere Antworten:

- [ ] a. IL and IRL are closely related since both recover the reward function
- [ ] b. IL and IRL are closely related since both learn good policies
- [x] c. IL and IRL are closely related since both learn from demonstrations
- [ ] d. IL and IRL are closely related since both learn a policy

Frage **8**

Vollständig

Erreichte Punkte 0,00 von 1,00

Given:

1. $\mathcal{L}(\cdot)$ is some loss function
2. $\breve{\pi}$ is the expert policy
3. $\mathcal{D}$ is the dataset of trajectories $\tau_j = \{(s_i, a_i)_{0:T}\}$ produced by $\breve{\pi}$
4. $p_{\text{data}}(s)$ is the distribution of states in $\mathcal{D}$
5. π_θ is the policy parametrized by θ
6. $\mathcal{D}_\theta$ is the dataset of trajectories $\tau_j = \{(s_i, a_i)_{0:T}\}$ produced by π_θ
7. $p_{\pi_\theta}(s)$ is the distribution of states in $\mathcal{D}_\theta$

Which of the following expressions correspond to the learning objective of **Behavioral Cloning**?

Wählen Sie eine oder mehrere Antworten:

- [ ] a. $\underset{\theta}{\operatorname{argmax}} \; \sum_{s \in \mathcal{D}} \mathcal{L}(\breve{\pi}(s_i), \pi_\theta(s_i))$
- [ ] b. $\underset{\theta}{\operatorname{argmax}} \; \mathrm{E}_{s \sim p_{\breve{\pi}}} \left[\mathcal{L}(\breve{\pi}(s), \pi_\theta(s))\right]$
- [ ] c. $\underset{\theta}{\operatorname{argmax}} \; \mathrm{E}_{s \sim p_{\text{data}}} \left[\mathcal{L}(\breve{\pi}(s), \pi_\theta(s))\right]$
- [x] d. $\underset{\theta}{\operatorname{argmin}} \; \mathrm{E}_{s \sim p_{\breve{\pi}}} \left[\mathcal{L}(\breve{\pi}(s), \pi_\theta(s))\right]$
- [ ] e. $\underset{\theta}{\operatorname{argmin}} \; \sum_{s \in \mathcal{D}_\theta} \mathcal{L}(\breve{\pi}(s_i), \pi_\theta(s_i))$
- [ ] f. $\underset{\theta}{\operatorname{argmin}} \; \mathrm{E}_{s \sim p_{\text{data}}} \left[\mathcal{L}(\text{ \breve{\pi}}(s), \pi_\theta(s))\right]$
- [ ] g. $\underset{\theta}{\operatorname{argmin}} \; \sum_{s \in \mathcal{D}} \mathcal{L}(\breve{\pi}(s_i), \pi_\theta(s_i))$
- [ ] h. $\underset{\theta}{\operatorname{argmax}} \; \sum_{s \in \mathcal{D}_\theta} \mathcal{L}(\breve{\pi}(s_i), \pi_\theta(s_i))$

Frage **9**

Vollständig

Erreichte Punkte 0,00 von 1,00

Given:

1. $\mathcal{L}(\cdot)$ is some loss function
2. π^* is the expert policy
3. $\mathcal{D}$ is the dataset of trajectories $\tau_j = \{(s_i, a_i)_{0:T}\}$ produced by π^*
4. $p_{\text{data}}(s)$ is the distribution of states in $\mathcal{D}$
5. π_θ is the policy parametrized by θ
6. $\mathcal{D}_\theta$ is the dataset of trajectories $\tau_j = \{(s_i, a_i)_{0:T}\}$ produced by π_θ
7. $p_{\pi_\theta}(s)$ is the distribution of states in $\mathcal{D}_\theta$

Which of the following expressions correspond to the learning objective of **General Imitation Learning**?

Wählen Sie eine oder mehrere Antworten:

- [ ] a. $\underset{\theta}{\operatorname{argmin}} \sum_{s \in D_\theta} \mathcal{L}(\pi^*(s_i), \pi_\theta(s_i))$
- [ ] b. $\underset{\theta}{\operatorname{argmin}} \mathrm{E}_{s \sim p_{\pi_\theta}} \left[\mathcal{L}(\pi^*(s), \pi_\theta(s))\right]$
- [ ] c. $\underset{\theta}{\operatorname{argmax}} \sum_{s \in D_\theta} \mathcal{L}(\pi^*(s_i), \pi_\theta(s_i))$
- [x] d. $\underset{\theta}{\operatorname{argmin}} \mathrm{E}_{s \sim p_{\text{data}}} \left[\mathcal{L}(\pi^*(s), \pi_\theta(s))\right]$
- [ ] e. $\underset{\theta}{\operatorname{argmax}} \mathrm{E}_{s \sim p_{\pi_\theta}} \left[\mathcal{L}(\pi^*(s), \pi_\theta(s))\right]$
- [ ] f. $\underset{\theta}{\operatorname{argmax}} \mathrm{E}_{s \sim p_{\text{data}}} \left[\mathcal{L}(\pi^*(s), \pi_\theta(s))\right]$
- [ ] g. $\underset{\theta}{\operatorname{argmin}} \sum_{s \in D} \mathcal{L}(\pi^*(s_i), \pi_\theta(s_i))$
- [ ] h. $\underset{\theta}{\operatorname{argmax}} \sum_{s \in D} \mathcal{L}(\pi^*(s_i), \pi_\theta(s_i))$

Frage **10**

Vollständig

Erreichte Punkte 0,00 von 1,00

Supervised Learning works under the assumption that both **training** and **test** data are independend and identically distributed (**i.i.d.**). What is the reason or reasons why this assumptions does not hold in Imitation Learning (IL)?

Wählen Sie eine oder mehrere Antworten:

- [ ] a. Because in IL, **training data is not i.i.d.** due to the Markov property
- [x] b. Because in IL, **test data is not i.i.d.** due to the Markov property
- [ ] c. Because in IL, both **training and test data are not i.i.d.** since future states depend on current actions
- [ ] d. Because in IL, **training data is not i.i.d.** since future states depend on current actions
- [ ] e. Because in IL, **test data is not i.i.d.** since future states depend on current actions
- [ ] f. Because in IL, both **training and test data are not i.i.d**. due to the Markov property

◀ [Exam] Inverse Reinforcement Learning

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